

AMENDMENT NO. 1 JANUARY 1989
TO
IS : 12003 - 1987 SPECIFICATION FOR
FENVALERATE, TECHNICAL

(Page 4, Table 1) — Insert the following matter after Sl No. (iii) under the respective columns:

(1)	(2)	(3)	(4)	(5)
iv)	3-phenoxybenzaldehyde (MPB) content, percent [by mass, <i>Max</i>	2.0	B	—

(Page 11) — Insert the following appendix after A-3.6:

‘ A P P E N D I X B

[Table 1, Item (iv)]

DETERMINATION OF 3-PHENOXYBENZALDEHYDE (MPB)

B-0. PRINCIPLE

B-0.1 A GIC unit with FI detector is used for this determination. Using solutions containing known amounts of the standard 3-phenoxybenzaldehyde (MPB), sample and the internal standard (*Is*), the response factor, *R*, for MPB in the (*Is*) approach is arrived at. A solution containing known mass of the fenvalerate sample (under investigation), and *Is* is injected subsequently into the GIC unit. The percentage of MPB in the sample is then computed by the standard relationship.

B-1. APPARATUS

B-1.1 Gas Liquid Chromatograph — Suitable for analysis when operated under the following suggestive operating conditions. These conditions can be varied provided standardization is done.

Column	: 1 800 mm × 6.3 mm glass column filled with 3 percent OV-225 coated on chromosorb WHP (80-100 mesh)
Temperature of column	: 210°C
Injection port	: 160°C
Detector	: 260°C
Carrier gas	: Nitrogen at 40 ml/min

B-1.2 Volumetric Flask

B-1.3 Microsyringe

B-2. REAGENTS

B-2.1 Metaphenoxy Benzaldehyde — AR Grade.

B-2.2 Chloroform — AR Grade.

B-2.3 Di-n-Butyl Phthalate — Internal standard (*Is*).

B-3. PREPARATION OF STANDARDS AND CALIBRATION

B-3.1 Weigh out accurately 2.5 g of di-n-butyl phthalate into a 50-ml volumetric flask and make up to volume with chloroform. This will give a solution containing 50 mg/ml of the *Is*.

B-3.2 Weigh out accurately 2.5 g of standard metaphenoxy benzaldehyde into a 50-ml volumetric flask and make up to volume with chloroform. This will give a solution containing 50 mg/ml of metaphenoxy benzaldehyde. Pipette out 1.0, 2.0, 3.0, 4.0 and 5.0 ml of this metaphenoxy benzaldehyde solution into a clean dry 10-ml flask. Add 5 ml of *Is* solution in each flask. Make up the volume with chloroform in the first four cases.

B-3.3 Inject 0.2 μ l of the standard solutions described under B-3.2 into the GLC unit. From the integrator print out and note down the peak areas of the metaphenoxy benzaldehyde and internal standard peaks.

B-3.4 Procedure

B-3.4.1 Weigh out accurately 5.0 g of the fenvalerate sample into a 10-ml volumetric flask. Measure out 5-ml of the *Is* solutions and make up to the mark with chloroform.

B-3.4.2 Introduce 0.2 μ l of this sample solution the GLC unit. From the integrator print out, note down the peak areas of metaphenoxy benzaldehyde and internal standard and internal standard peaks. Compute the percentage of the metaphenoxy benzaldehyde content in the sample.

B-3.5 Calculation

$$\text{3-phenoxybenzaldehyde content, percent by mass} = \frac{R \times A_1 \times m_1}{A_s \times m_s} \times 100$$

where

R = response factor,

A_1 = area of the 3-phenoxybenzaldehyde peak obtained while analyzing the sample,

m_1 = mass of the internal standard added,

A_s = area of the internal standard peak, and

m_s = mass of the sample taken for analysis.'

(AFGDC 6)